

Python

Basic String

using placeholder

```
x = 10
s = "Hello world %d"%x
```

using formatting

```
x = 12
s = f"Hello world {x}"
```

Logging

```
import logging
import numpy as np

logging.basicConfig(filename='logfile.log',
                    format='%(asctime)s %(levelname)s: %(message)s',
                    filemode='w')

logger = logging.getLogger()
logger.setLevel(logging.DEBUG)
x=np.random.random((2,3,4))
logger.debug(f'x :{x}')
logger.info("Just an information")
logger.warning("Its a warning")
logger.error("Did you try to divide by zero?")
logger.critical("Internet is down")
```

Creating a matrix

```
mat1 = [[0]*n_cols]*n_rows
# or
mat2 = [[0]*n_cols for _ in range(n_rows)]
```

Numpy

np.random.rand(2,3,4) -> standard uniform

np.random.random((2,3)) -> random_sample from standard uniform

np.random.uniform(0,5,(2,3)) -> non-standard uniform distribution

np.random.randn(2,3,4) -> standard normal

np.random.normal(0,5,(2,3)) -> non-standard normal distribution

```
$a1 = np.random.rand(5,10,2)
$b1 = np.random.rand(5,5,2)
$np.hstack((a1,b1)).shape
(5,15,2)
$np.concatenate((a1,b1),axis=1)
a2 = np.random.rand(10,5,2)
b2 = np.random.rand(5,5,2)
$np.vstack((a2,b2)).shape
(15,5,2)
$np.concatenate((a2,b2),axis=0).shape
(15,5,2)
```

np.nonzero(J) -> returns tuples of indices of J that are nonzero

np.maximum(x1, x2) -> returns the maximum of x1 and x2, element-wise. This is a scalar if both x1 and x2 are scalars.

np.max(J, axis=1) -> Maximum of a. If axis is None, the result is a scalar value. If axis is an int, the result is an array of dimension a.ndim - 1. If axis is a tuple, the result is an array of dimension a.ndim - len(axis).

np.where(J>t, J, 0) -> fills the position where the boolean condition is true with elements of J, 0 otherwise.

Get indices of N maximum values in a 2D array: example: used in NMS in Harris corner detection.

```
n=5
arr=np.random.rand(2,3)
x, y = np.unravel_index(np.argsort(arr.ravel(), axis=None)[::-1][:n], arr.shape)
```

Creating a grid using outer product and np.meshgrid for grid indices

Code:

```
# Create a grid-like structure using outer product.
# Used in the implementation of positional encoding
# in Transformers.
a = np.arange(5)
b = np.arange(5)
c = np.arange(5, 0, -1)

grid1 = a[:, np.newaxis] @ b[np.newaxis, :]
grid2 = a[:, np.newaxis] @ c[np.newaxis, :]
print(grid1)
print(grid2)

x, y = np.meshgrid(a, b, indexing='ij')
for (i, j) in zip(x, y):
    print("i: ", i, ", j: ", j)
```

Output:

```
[[ 0  0  0  0  0]
 [ 0  1  2  3  4]
 [ 0  2  4  6  8]
 [ 0  3  6  9 12]
 [ 0  4  8 12 16]]
[[ 0  0  0  0  0]
 [ 5  4  3  2  1]
 [10  8  6  4  2]
 [15 12  9  6  3]
 [20 16 12  8  4]]
i: [0 0 0 0 0], j: [0 1 2 3 4]
i: [1 1 1 1 1], j: [0 1 2 3 4]
i: [2 2 2 2 2], j: [0 1 2 3 4]
i: [3 3 3 3 3], j: [0 1 2 3 4]
i: [4 4 4 4 4], j: [0 1 2 3 4]
```

Slicing and Indexing in Numpy

Example: numpy_slicing_and_indexing.py

1. Code:

```
# -*- coding: utf-8 -*-
"""
Created on Tue Mar 31 15:12:13 2026

@author: reina
"""

import numpy as np

x = np.random.randint(0, 10, (3, 4, 5))
print(x.shape)
print(x)
print('\n')
# l = [i for i in x]
print([i for i in x]) # List comprehension
print('\n')
# The order returned in list comprehension is
# consistent with the order of priority of indexing
# of arrays in numpy.
print(x[0, :])
print(x[0, :, :])
print('\n')
print(x[1, :])
print(x[1, ...])
print('\n')

print((x[0, :] == x[0, :, :]).all())
print((x[1, :] == x[1, ...]).all())
```

Output:

```
(3, 4, 5)
[[[3 6 7 2 3]
  [0 6 4 5 5]
  [9 4 5 2 0]
  [7 2 4 3 9]]

 [[5 9 9 7 3]
  [9 4 8 4 6]
  [0 7 0 6 6]
  [5 3 9 0 8]]]
```

```

[[1 5 7 9 2]
 [5 2 9 4 2]
 [4 8 6 4 2]
 [3 2 7 8 3]]

[array([[3, 6, 7, 2, 3],
        [0, 6, 4, 5, 5],
        [9, 4, 5, 2, 0],
        [7, 2, 4, 3, 9]], dtype=int32), array([[5, 9, 9, 7, 3],
        [9, 4, 8, 4, 6],
        [0, 7, 0, 6, 6],
        [5, 3, 9, 0, 8]], dtype=int32), array([[1, 5, 7, 9, 2],
        [5, 2, 9, 4, 2],
        [4, 8, 6, 4, 2],
        [3, 2, 7, 8, 3]], dtype=int32)]

[[[3 6 7 2 3]
 [0 6 4 5 5]
 [9 4 5 2 0]
 [7 2 4 3 9]]
 [[3 6 7 2 3]
 [0 6 4 5 5]
 [9 4 5 2 0]
 [7 2 4 3 9]]]

[[[5 9 9 7 3]
 [9 4 8 4 6]
 [0 7 0 6 6]
 [5 3 9 0 8]]
 [[5 9 9 7 3]
 [9 4 8 4 6]
 [0 7 0 6 6]
 [5 3 9 0 8]]]

True
True

```

Slicing and Indexing in Python

1. Code: slicing_and_indexing.py

```

"""
Slicing in Python:
    a:b:c
a: start
b: end
c: step
"""

a = [1, 2, 3]
b = a
print(a == b)
print(a is b)

c = [1, 2, 3]
d = c.copy()
# Copy a list using :: operator
e = c[::]
print(c == d)
print(c is d)
print(c is e)

# Reverse a list
print(c[::-1])

```

Out:

```

True
True
True
False
False
[3, 2, 1]

```

Matplotlib

1. fig.add_subplot

```

# Plot the images in the batch, along with the corresponding labels
fig = plt.figure(figsize=(10, 4))
# Display 20 images
for idx in np.arange(27):
    ax = fig.add_subplot(3, 9, idx + 1, xticks=[], yticks=[])

```

```
    imshow(example_data[idx].detach().numpy())
    ax.set_title(classes[example_targets[idx]])
plt.tight_layout()
plt.show()
```

2. plt.subplots

```
# Plot the images in the batch, along with the corresponding labels
fig, subs = plt.subplots(2, 10, figsize=(25, 4))
for idx, sub in zip(np.arange(20), subs.flatten()):
    sub.imshow(np.squeeze(images[idx]), cmap="gray")
    sub.set_title(str(labels[idx].item()))
    sub.axis("off")
plt.savefig("MNIST-data.png")
plt.show()
```